

Westhaven Cancer Institute

Breast Pathology and Molecular Diagnostics

Patient:	[REDACTED, A.] (MRN 29103)
Age / Sex:	52 / F
Accession:	WCI-26-004115
Procedure date:	2026-04-22
Reported:	2026-04-29

FINAL SURGICAL PATHOLOGY DIAGNOSIS

A. Left breast, partial mastectomy: Invasive breast carcinoma of no special type, Nottingham grade 2, ER+/PR+/HER2-negative, PIK3CA-mutant; 2.1 cm; closest margin 4 mm (anterior); no LVI; associated intermediate-grade DCIS (~10%).

B. Sentinel lymph nodes (3): negative for metastatic carcinoma (0/3).

SYNOPTIC SUMMARY (CAP-style)

Element	Result
Procedure	Partial mastectomy with sentinel lymph node biopsy
Tumor site	Left breast, upper outer quadrant
Tumor size	2.1 cm (largest invasive focus)
Histologic type	Invasive carcinoma, no special type (ductal)
Nottingham score	3 + 2 + 2 = 7/9 → Grade 2
Lymphovascular invasion	Not identified
DCIS component	Present, intermediate grade, solid pattern, ~10%
Margin status	Negative; closest 4 mm anteriorly
Lymph node status	0 / 3 sentinel nodes positive
Pathologic stage	pT2 N0 (AJCC 8th)

CLINICAL HISTORY

52 y.o. postmenopausal F. Screening mammography: 2.1 cm spiculated mass, upper outer quadrant, left breast. Core needle bx: invasive carcinoma. Underwent partial mastectomy + SLNB on 2026-04-22. No personal hx of breast cancer. Mother dx'd breast cancer at age 68. No other significant FHx.

GROSS DESCRIPTION

A. Fibrofatty breast tissue 7.5 x 6.0 x 3.1 cm. A firm, spiculated, gray-white tumor measuring 2.1 x 1.9 x 1.6 cm in the upper outer aspect. Closest margin 4 mm anteriorly. B. Three sentinel lymph nodes; largest 1.4 cm.

MICROSCOPIC DESCRIPTION

A. Invasive carcinoma of no special type, composed of nests and cords of moderately atypical epithelial cells infiltrating fibrous stroma. Nottingham grade 2 (tubule formation 3 + nuclear pleomorphism 2 + mitoses 2 = 7/9). No lymphovascular invasion identified. Margins negative; closest margin 4 mm anteriorly. Associated DCIS, intermediate grade, solid pattern, involving ~10% of the tumor. B. All three sentinel lymph nodes negative for metastatic carcinoma (0/3) by H&E; and pankeratin IHC.

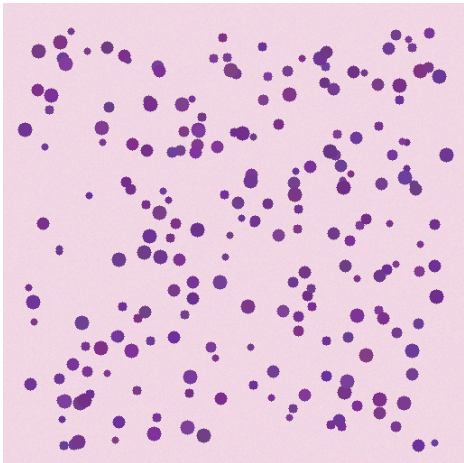
BIOMARKER STUDIES

Marker	Result	Score / Notes
ER	Positive (95% strong nuclear)	Allred 8/8
PR	Positive (40% strong nuclear)	Allred 7/8
HER2 (4B5)	1+	Negative; no FISH indicated
Ki-67 (MIB-1)	Approximately 18%	Intermediate proliferation index

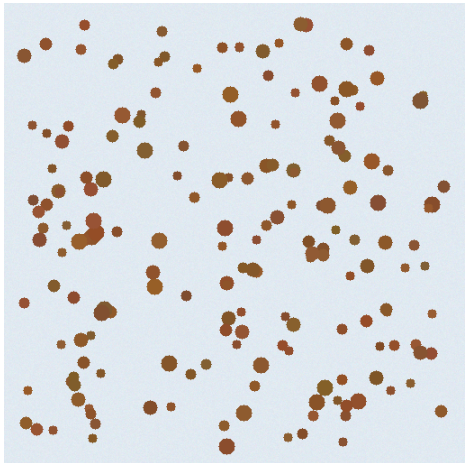
Ancillary IHC:

- **E-cadherin:** Retained (membranous)
- **AE1/AE3:** Positive in tumor and sentinel-node H&E; re-section

REPRESENTATIVE PHOTOMICROGRAPHS



H&E; ×20 — invasive ductal carcinoma NST



ER IHC ×20 — Allred 8/8

MOLECULAR PROFILING (companion diagnostics)

Somatic sequence variants

Gene	HGVSc	HGVSp	VAF	Interpretation
PIK3CA	c.3140A>G	p.H1047R	0.32	Pathogenic — Actionable per NCCN BRST-V (2026)

Copy-number alterations of interest:

- **ERBB2** — copy number 2 (no amplification (consistent with HER2 IHC 1+))

Microsatellite Instability:	MSS
TMB:	3.6 mut/Mb
Germline screen:	BRCA1 / BRCA2 / PALB2 / TP53 (germline panel): no pathogenic or likely pathogenic variants

RECURRENCE-RISK GENOMIC ASSAY

Assay: 70-gene risk profile (in-house surrogate; not MammaPrint(TM))

Result: Low Risk (continuous index 0.18; binary cutoff 0.31)

Risk index below cutoff indicates lower 10-year distant-recurrence risk; in combination with classical clinicopathologic features, this may influence adjuvant chemotherapy decision-making per institutional guidelines.

INTERPRETATION / COMMENT

Invasive ductal carcinoma NST, Nottingham grade 2, 2.1 cm, ER+/PR+/HER2-negative. PIK3CA H1047R is somatic and may inform alpelisib + endocrine combinations in advanced/metastatic settings per current NCCN. 70-gene risk profile is Low Risk. Margins negative; 0/3 sentinel nodes positive. Adjuvant endocrine therapy is appropriate; chemotherapy decision per tumor board, informed by Low-Risk genomic result.

Electronically signed by **Dr. T. Okafor, MD**, Breast Pathology (2026-04-29)
Co-reviewed: Dr. H. Liu, MD (Molecular Pathology Director)

ADDENDUM (2026-05-06)

Outside germline panel result received: **NEGATIVE** for pathogenic variants in BRCA1, BRCA2, PALB2, CHEK2, ATM, TP53. Result added to Molecular Profiling section.

Footnotes: Allred score combines proportion + intensity (0-8). HER2 IHC scoring per ASCO/CAP 2018; 0/1+ negative, 2+ requires FISH, 3+ positive. 70-gene risk profile is a high-complexity in-house surrogate; results should be interpreted in clinical context.